

Natural Language Processing: Introduction

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09-Sep-2026

Bursting the Bubbles

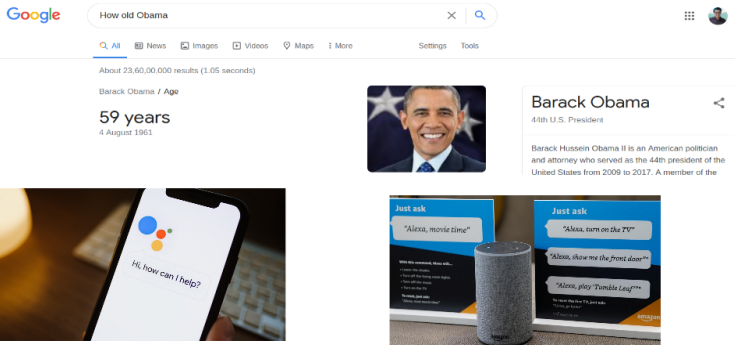
This session will try to discuss the terminologies from a non-technical perspective

- NLP
- Computational Linguistics
- Language Models
- LLMs
- ChatGPT
- Generative models

Acknowledgement: This presentation is compiled based on the inputs from various web resources and Text Books (SLP by Jurafsky and FSNLP by C D Manning)

- What is NLP?
- Stages in NLP
- Challenges
- Applications

Let's start with some intuition



The screenshot shows a Google search interface. The search bar contains the text "How old Obama". Below the search bar, the results show "About 23,60,00,000 results (1.05 seconds)". A knowledge panel for Barack Obama is displayed on the right, showing his age as "59 years" and his birth date as "4 August 1961". Below the knowledge panel, there are two images: a smartphone displaying the Google Assistant interface with the text "Hi, how can I help?" and a smart speaker (Amazon Echo) next to a sign that lists various voice commands for Alexa, such as "Alexa, movie time" and "Alexa, turn on the TV".

Let's start with some intuition

English – detected

↔

Tamil

How are you



Both houses of the Indian Parliament use the Simultaneous Interpretation System for instant translation of a speech and its simultaneous delivery. The system was initially used for translating from Hindi to English and vice-versa in 1964 (for both houses). Currently, this system has been expanded for translation across the following languages - Lok Sabha: Kannada, Marathi, Tamil, Punjabi, Malayalam, Manipuri, Sanskrit, Telugu and Oriya Rajya Sabha: Assamese, Bengali, Tamil, Telugu, Kannada, Gujarati, Malayalam, Marathi, Urdu, Oriya

எப்படி இருக்கிறீர்கள்

Er

Malayalam

എന്തൊക്കെയാണ്

Hindi

क्या हाल है

kya haal hai





Can you create a quote on NLP



"Natural Language Processing weaves the tapestry of human expression into the fabric of computation, bridging the chasm between words and understanding with the threads of algorithms and meaning."



Who told this?



I generated that quote myself based on your request for a quote on NLP. It is not attributed to any specific individual. If you're looking for quotes from notable figures in the field of NLP, I can certainly provide you with some well-known quotes by experts. Just let me know!



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- Machine Learning
- Computational Intelligence

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 - Beating world champions in Chess/GO
- How an automated customer care chatbot understand 5 year old child?
- Understanding of human language is important in all the above application

What is NLP?

- **Natural Language Processing**
- Deals with building computational algorithms to **understand** and **generate** and **process** human languages
- Most of the information is represented in natural languages
- Applications of NLP are everywhere- web search, advertisement, emails, customer service, language translation, health, etc
- NLP is multidisciplinary
 - Computer Science
 - Linguistics
 - Statistics
 - Expert in specific language
 - Psycho-linguistics
 - Socio-linguistics

Where we are?

- Spell checker
- Part-of-speech tagging
- Spam detection
- Mail categorization
- Machine Translation
- Information retrieval
- Sentiment analysis
- Chatbots

- **Computational Linguistics provide automated solutions for the NLP problems**
- Computational linguistics is the application of linguistic theories and computational techniques to problems of NLP.
- Computational linguists are interested in providing **computational models** of various kinds of linguistic phenomena.
- Linguistically competent computers would greatly facilitate our interaction with machines and software of all sorts
- “Human knowledge is expressed in language. So computational linguistics is very important.” –Mark Steedman, ACL Presidential Address (2007)

What Computational Linguistics can do?

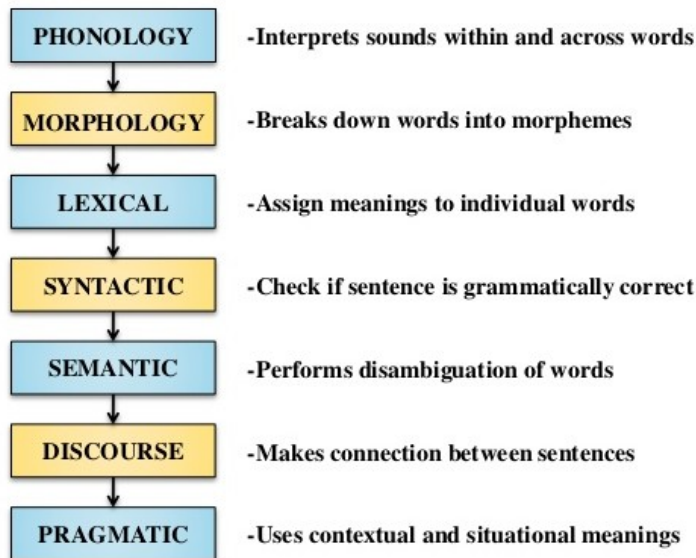
- CL can automate solutions for understanding language phenomena
 - Explain why 'ly' in ally is different from quickly
 - Tell the difference between '**water** the flowers' and 'drink the **water**'
 - Why 'ate *iddaly with chatni*' and 'ate iddaly with brother' are different
 - Translate 'how are you' to 'Sughamano' in Malayalam
- Understanding some basics of NLP will help to know CL better

- **Ultimate Goal:** Natural human-to-computer communication
 - Very long way to go...
- **Engineering Goal:** Build sophisticated applications based on the useful linguistic patterns
 - More practical and we are developing in this way
 - Not perfect, still you can find useful applications with acceptable errors

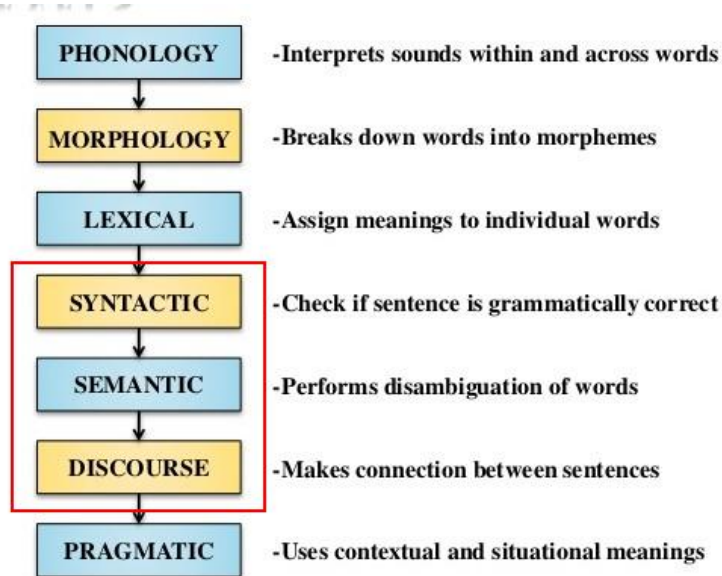
Why NLP is Important

- This is the age of data driven application
- Majority of this data is available in the form of Text
 - Social media posts, Wikipedia articles, News, stock market, research articles, clinical reports,...
- Mining these data for useful applications
 - Knowledge of the language is essential
- NLP deals with developing applications to process/understand these textual data automatically

Levels of Language Processing



Levels of NLP



- Context Sensitive
 - Human communication involve many forms of signals- expressions, or just silence
 - Understanding the intent of the communication

- Context Sensitive

- Human communication involve many forms of signals- expressions, or just silence
- Understanding the intent of the communication
- The dog could not cross the street, it was tired.
The dog could not cross the street, it was congested.

Ambiguity is everywhere
Lexical Level

Structure Level

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Lexical Level

- Rose rose to put rose roes on her rows of roses

Structure Level

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Lexical Level

- Rose rose to put rose roes on her rows of roses
 - Rose *[a person]* rose *[stood]* to put rose *[pink-colored]* roes *[fish eggs as fertilizer]* on her rows *[line]* of roses *[flower]*

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Lexical Level

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Structure Level

- The professor said on Monday he would give an exam.
- I ate Dosa with Chatni
I ate Dosa with brother
- The boy saw a man with Telescope.

Challenges

Idioms are challenging

the old girl finally
kicked the
bucket

×

വൃദ്ധ പെൺകുട്ടി ഒടുവിൽ
ബക്കറ്റ് തട്ടി

vṛd'dha peṇkuṭṭi oṭuvil bakkarr̥ taṭṭi

बूढ़ी लड़की ने आखिरकार बाल्टी
को लात मार दी

boodhee ladakee ne aakhirakaar
baaltee ko laat maar dee

Tamil

வயதான பெண்
இறுதியாக வாளியை
உதைத்தார்

Vayatāṇa peṇ iṟutiyāka vāliyai utaittār

- Languages are imprecise and vague
 - It is very *hot* here
 - Can the dog swim? (yes/no question)
 - Can you please post this letter? (request)

- Non-categorical nature of languages
 - The meaning of words are not categorical
 - Meaning and usages are changing time to time
 - Eg: *while* earlier used with meaning of time (take a while), now a complementizer introducing subordinate clause (...., while he was playing)
 - Same word may have different part of speech (He *saw* a boy; the *saw* was sharp
 - There are gradual change in usages
- How these challenges can be explained?

Two Approaches

- Rule-based Approaches

- We have (large) set of linguistic rules to explain all phenomena
- There are exceptions which cannot be explained by these general rules
- Not scalable with respect to the evolving nature of languages

- Empirical Approach

- Observe a large collection (corpus) to make inferences
- Use statistical analysis to explain language events
- More reliable, provided, sufficient amount of clean data to learn

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- Predict the next word given a set of previous word

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- Train the language models for specific task- say translation, sentiment analysis

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- Once such a model is ready we can **fine-tune** them for specific applications
- Two major classes of LLMs: Encoders/Abstractive models, Generative models (like ChatGPT)